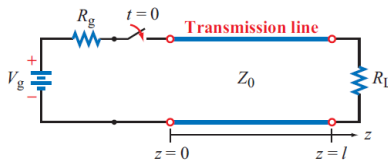


Transients Refresher



Let's make that more general



Generally, all the following could happen:

- Mismatch between Generator and line
 - If $R_g \neq Z_0$
- Mismatch between line and load
 - If $R_L \neq Z_0$
- Line length will cause propagation delay
 - $T = \frac{l}{v_{ph}}$

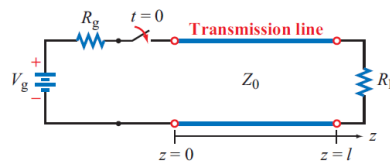
All these phenomena will happen at different location of the line and different time stamps

1

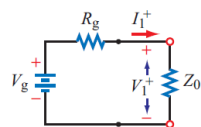
1

Transients

*The **transient response** of a voltage pulse on a transmission line is a time record of its back and forth travel between the sending and receiving ends of the line, taking into account all the multiple reflections (echoes) at both ends.*



(a) Transmission-line circuit



(b) Equivalent circuit at $t = 0^+$

Goal is to get $V(z, t)$ when switch is close
That is called Step response

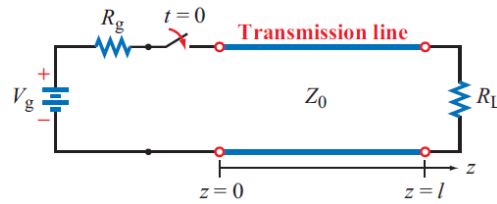
2

Transient Response

Initial current and voltage (at $t = 0^+$)

$$V_1^+ = I_1^+ Z_0 = \frac{V_g Z_0}{R_g + Z_0}$$

Using Voltage Divider in Fig (b)



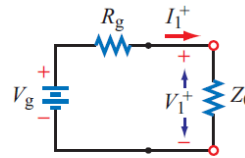
(a) Transmission-line circuit

Reflection at the load

$$V_1^- = \Gamma_L V_1^+,$$

Load reflection coefficient

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$



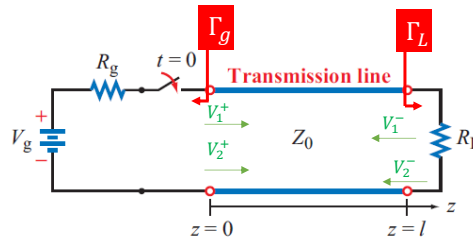
(b) Equivalent circuit at $t = 0^+$

Second transient (at $t = 2T$)

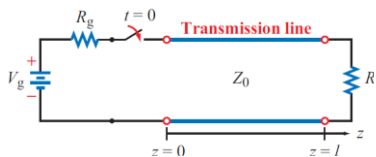
$$V_2^+ = \Gamma_g V_1^- = \Gamma_g \Gamma_L V_1^+$$

Generator reflection coefficient

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0}$$



3



$$R_g = 4Z_0 \text{ and } R_L = 2Z_0.$$

Voltage Wave

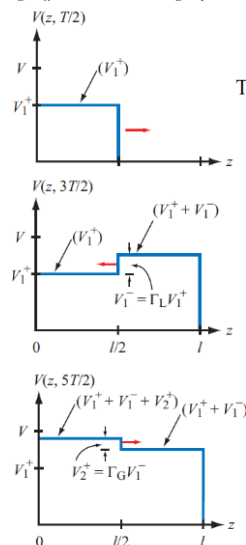
$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0}$$

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$

The corresponding reflection coefficients are $\Gamma_L = 1/3$ and $\Gamma_g = 3/5$.

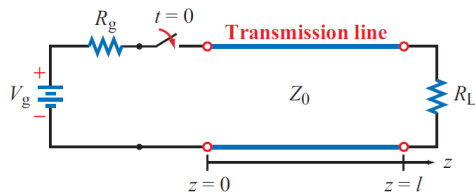
$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}$$

$T = l/v_{ph}$ is the time it takes the wave to travel the full length of the line



4

Steady State Response

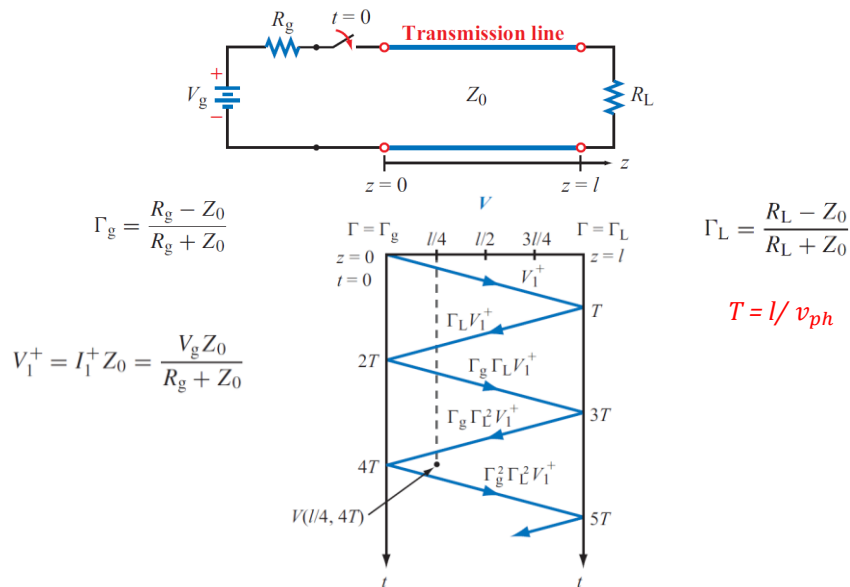


(a) Transmission-line circuit

$$V_{\infty} = \frac{V_g R_L}{R_g + R_L}$$

The multiple-reflection process continues indefinitely, and the ultimate value that $V(z, t)$ reaches as t approaches $+\infty$ is the same at all locations on the transmission line.

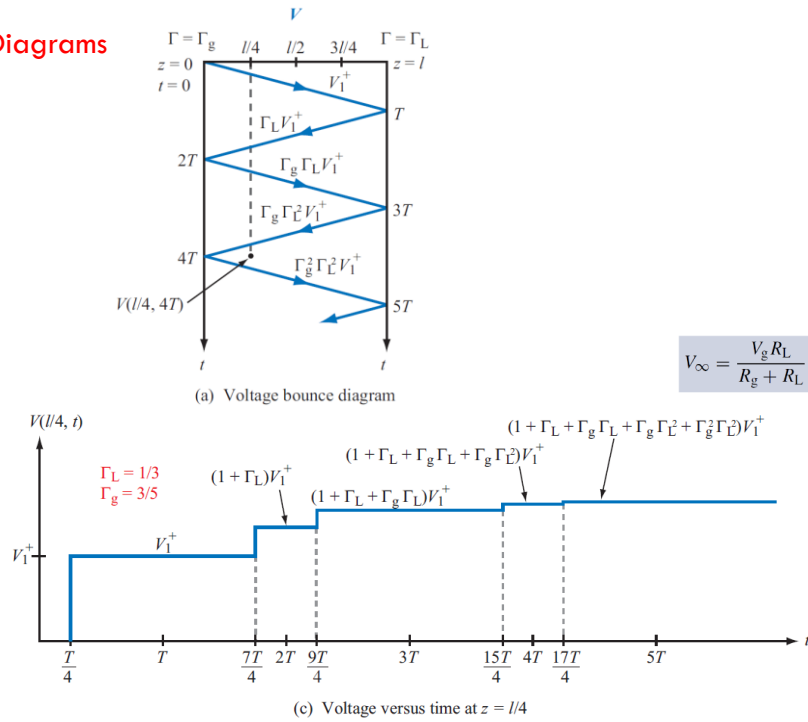
5



6

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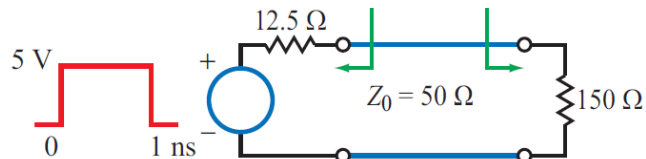
Bounce Diagrams



7

Example 2-15: Pulse Propagation

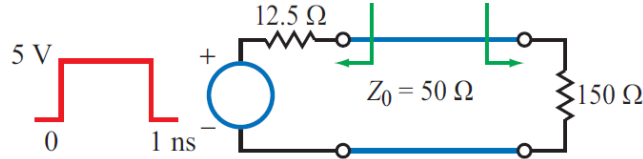
The transmission-line circuit of Fig. 2-43(a) is excited by a rectangular pulse of duration $\tau = 1$ ns that starts at $t = 0$. Establish the waveform of the voltage response at the load, given that the pulse amplitude is 5 V, the phase velocity is c , and the length of the line is 0.6 m.



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Example 2-15: Pulse Propagation

The transmission-line circuit of Fig. 2-43(a) is excited by a rectangular pulse of duration $\tau = 1$ ns that starts at $t = 0$. Establish the waveform of the voltage response at the load, given that the pulse amplitude is 5 V, the phase velocity is c , and the length of the line is 0.6 m.



$$T = \frac{l}{c} = \frac{0.6}{3 \times 10^8} = 2 \text{ ns.}$$

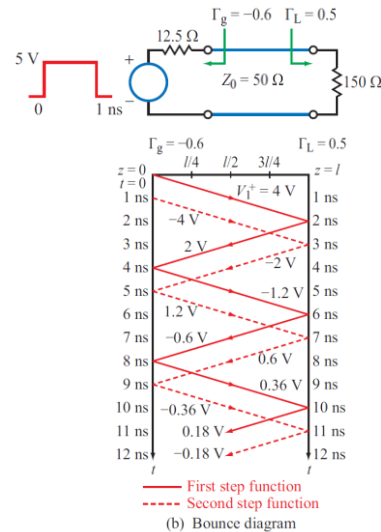
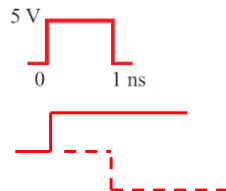
$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0} = \frac{150 - 50}{150 + 50} = 0.5$$

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0} = \frac{12.5 - 50}{12.5 + 50} = -0.6.$$

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By Eq. (2.147), the pulse is treated as the sum of two step functions, one that starts at $t = 0$ with an amplitude $V_{10} = 5$ V and a second one that starts at $t = 1$ ns with an amplitude $V_{20} = -5$ V. Except for the time delay of 1 ns and the sign reversal of all voltage values, the two step functions will generate identical bounce diagrams, as shown in Fig. 2-43(b). For the first step function, the initial voltage is given by

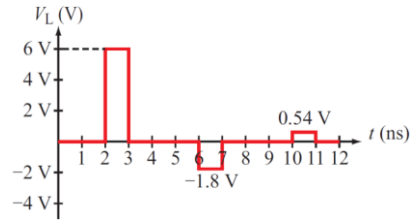
$$V_1^+ = \frac{V_{01} Z_0}{R_g + Z_0} = \frac{5 \times 50}{12.5 + 50} = 4 \text{ V.}$$



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Example 2-15: Pulse Propagation

The transmission-line circuit of Fig. 2-43(a) is excited by a rectangular pulse of duration $\tau = 1$ ns that starts at $t = 0$. Establish the waveform of the voltage response at the load, given that the pulse amplitude is 5 V, the phase velocity is c , and the length of the line is 0.6 m.



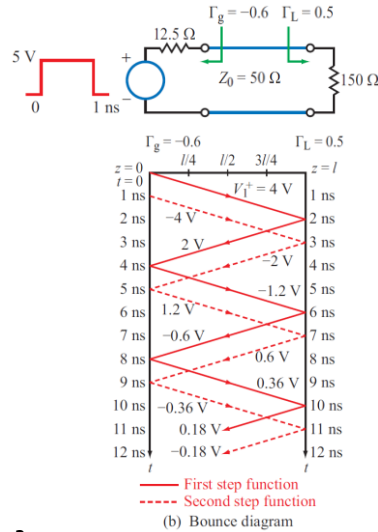
(c) Voltage waveform at the load

What do we do if we want the voltage across the load with time?

To do this we move down the right-hand side of the bounce diagram, and count voltage waves as we move across them.

We start out at zero, of course, and do not see anything until we get to 2 ns. Then we cross the 4V (V_1^+) wave and we cross the 2V (V_1^-) wave at the same time of the first step function, so the voltage only goes up to +6V.

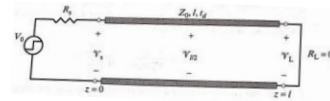
At 3 ns, we cross -4V (V_1^+) and we cross the -2V (V_1^-) wave at the same time of the second step function, so the voltage only goes down by -6V, sum is 0V.



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Example 2: Short circuited line

A transmission line with characteristics impedance of Z_0 of length l is terminated with a short circuit on the end as shown in the figure. Calculate the voltage V_s as a function of time. Let us calculate the values for $Z_s = Z_0/3$; $Z_s = Z_0$; and $Z_s = 3Z_0$.

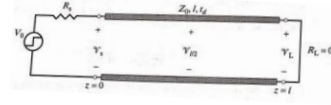


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Example 2: Short circuited line

A transmission line with characteristics impedance of Z_0 of length l is terminated with a short circuit on the end as shown in the figure. Calculate the voltage V_s as a function of time. Let us calculate the values for $Z_s = Z_0/3$; $Z_s = Z_0$; and $Z_s = 3Z_0$.



$$\Gamma_L = \frac{0 - Z_0}{0 + Z_0} = -1$$

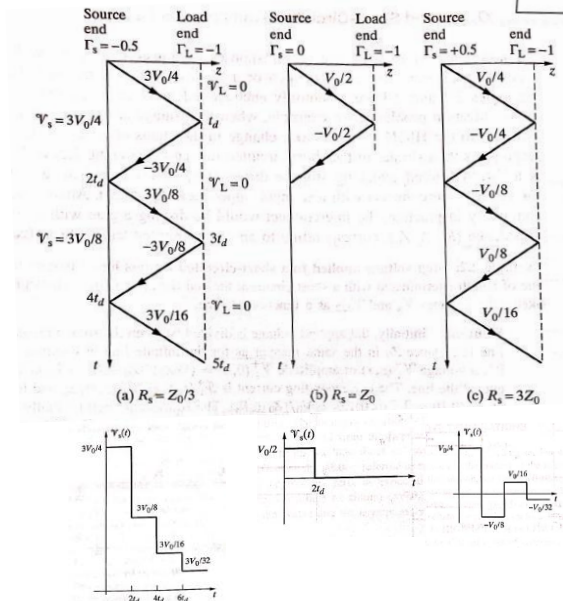
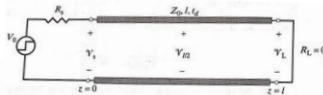
$\Gamma_s = -0.5, 0$, and 0.5 for the three source impedances respectively

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}$$

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Example 2: Short circuited line

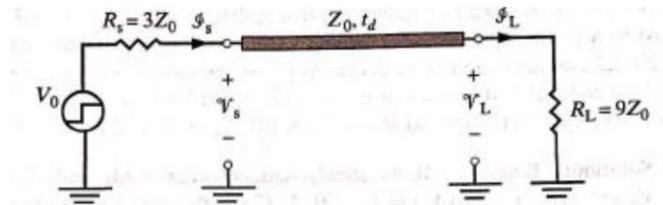


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Example 3: Step response of resistively terminated line

Consider the circuit shown below. Sketch V_L , and V_S as a function of time

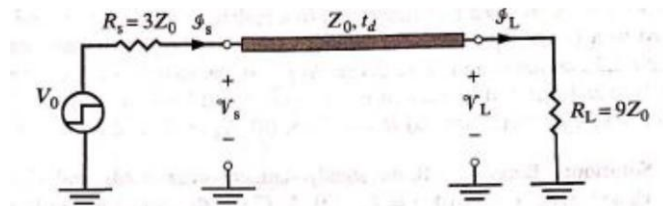


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Example 3: Step response of resistively terminated line

Consider the circuit shown below. Sketch V_L , and V_S as a function of time



In the first step we calculate Γ_L and Γ_S . As a habit, I put the values on the Bounce diagrams themselves. Then we calculate, the amplitude of V_1^+ as $\frac{Z_0 V_0}{Z_0 + Z_S}$ which in this case will be $0.25 V_0$.

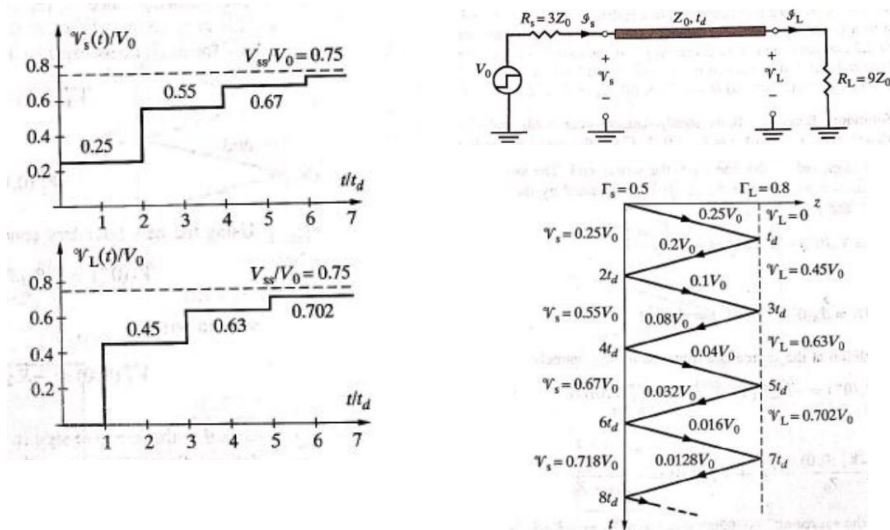
$$\Gamma_S = 0.5 \quad \Gamma_L = 0.8$$

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Example 3: Step response of resistively terminated line

Consider the circuit shown below. Sketch V_L and V_s as a function of time

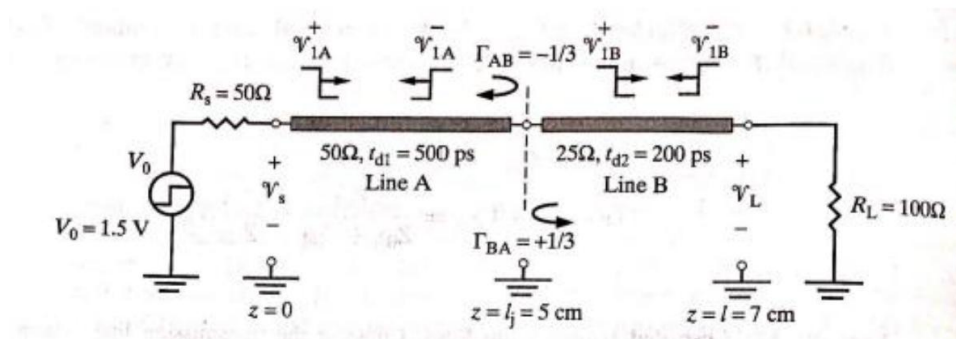


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Example 4: Cascaded Transmission Lines

Consider the circuit shown below. Sketch V_L and V_s as a function of time

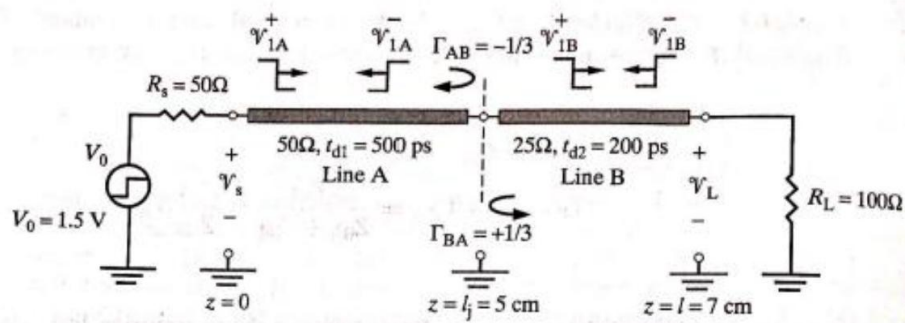


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Example 4: Cascaded Transmission Lines

Consider the circuit shown below. Sketch V_L and V_s as a function of time



We calculate Γ_L , and Γ_s . The amplitude of V_{1A}^+ is calculated as $\frac{Z_{0A}V_0}{Z_{0A}+Z_s}$. In addition we need to calculate the reflection coefficient from transmission line A to B and also B to A. They will just be negative of each other. The various coefficients are also shown in the figure. Once we have the coefficients, we can start drawing the bounce diagram. We have to take care that waves will be transmitted from A to B and from B to A. Once we have the diagram, we have to just add the waves as they arrive to get voltage curves. Bounce diagrams and the voltage curves are shown in the figure below.

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Example 4: Cascaded Transmission Lines

Consider the circuit shown below. Sketch V_L and V_s as a function of time

$$\Gamma_{AB} = \frac{25 - 50}{25 + 50} = -1/3$$

$$\Gamma_{BA} = -\Gamma_{AB} = 1/3$$

$$\tau_{AB} = 1 + \Gamma_{AB} = 2/3$$

$$\tau_{BA} = 1 + \Gamma_{BA} = 4/3$$

$$V_{1A}^+ = \frac{V_0 Z_{0A}}{Z_{0A} + R_s} = 0.75V$$

$$V_{1A}^- = V_{1A}^+ \Gamma_{AB} = -0.25V$$

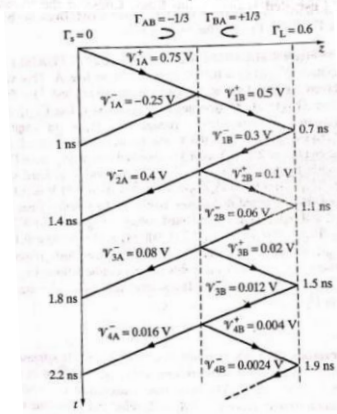
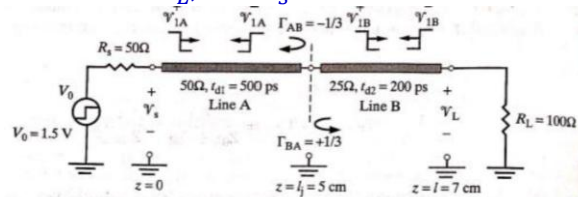
$$V_{1B}^+ = V_{1A}^+ \tau_{AB} = 0.5V$$

$$V_{1B}^- = V_{1B}^+ \Gamma_L = 0.3V$$

$$V_{2A}^- = V_{1B}^- \Gamma_{BA} = 0.4V$$

$$V_{2B}^+ = V_{1B}^- \Gamma_{BA} = 0.1V$$

$$V_{2B}^- = V_{2B}^+ \Gamma_L = 0.06V$$

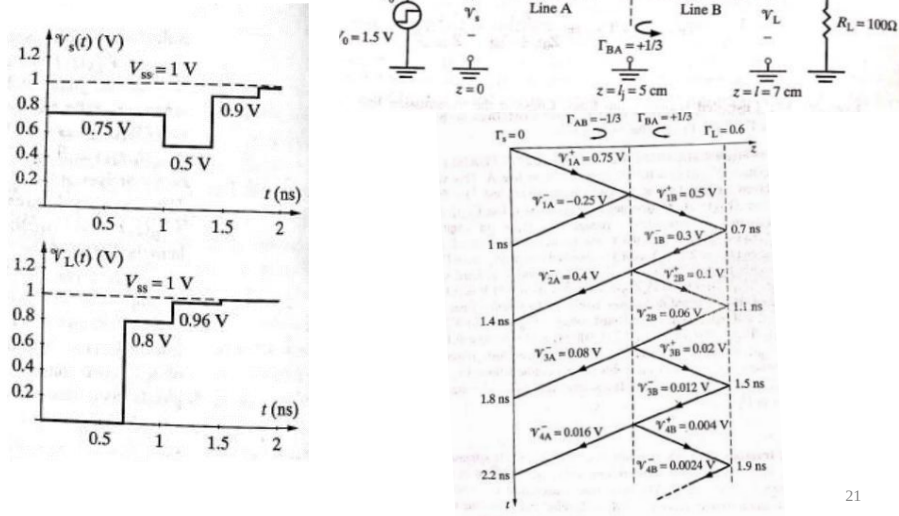


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Example 4: Cascaded Transmission Lines

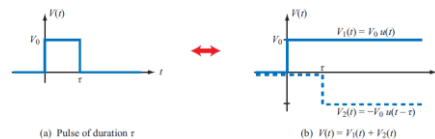
Consider the circuit shown below. Sketch V_L and V_S as a function of time



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Summary



Rectangular pulse is equivalent to the sum of two step functions

Chapter 2 Relationships

TEM Transmission Lines

$$L'C' = \mu\epsilon$$

$$\frac{G'}{C'} = \frac{\sigma}{\varepsilon}$$

$$\alpha = \Re(\gamma) = \Re\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{Np/m})$$

$$\beta = \Im m(\gamma) = \Im m\left(\sqrt{(R' + j\omega L')(G' + j\omega C')}\right) \quad (\text{rad/m})$$

$$Z_0 = \frac{R' + j\omega L'}{\gamma} = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}} \quad (\Omega)$$

$$\Gamma = \frac{z_L - 1}{z_L + 1}$$

Lossless Line

$\alpha = 0$

$$\beta = \omega \sqrt{L'C'}$$

$$Z_0 = \sqrt{\frac{L'}{C'}}$$

$$u_p = \frac{1}{\sqrt{\mu\epsilon}} \quad (\text{m/s})$$

$$\lambda = \frac{u_p}{f} = \frac{c}{f} \frac{1}{\sqrt{\epsilon_r}} = \frac{\lambda_0}{\sqrt{\epsilon_r}}$$

$$d_{\max} = \frac{\theta_r \lambda}{4\pi} + \frac{n\lambda}{2}$$

Step Function Transient Response

$$V_1^+ = \frac{V_g Z_0}{R_g + Z_0}$$

$$V_{\infty} = \frac{V_g R_L}{R_g + R_L}$$

$$\Gamma_g = \frac{R_g - Z_0}{R_g + Z_0}$$

$$\Gamma_L = \frac{R_L - Z_0}{R_L + Z_0}$$

$$d_{\min} = \frac{\theta_r \lambda}{4\pi} + \frac{(2n+1)\lambda}{4}$$

$$S = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

$$P_{av} = \frac{|V_0^+|^2}{2Z_0} [1 - |\Gamma|^2]$$

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